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Given mixture contains 15.47 lit of methane and 0.213 lit of propane, hence

$$\Delta H_C = \left[\left(\frac{\Delta H_C\ (CH_4)}{22.4\ \text{lit}}\right) \text{x lit}\right] + \left[\left(\frac{\Delta H_C\ (C_3H_8)}{22.4\ \text{lit}}\right)(15.68 - \text{x})\ \text{lit}\right]$$

$$\Delta H_C = \left[\left(\frac{-890\ \text{kJ mol}^{-1}}{22.4\ \text{lit}}\right) 15.47 \quad \text{lit}\right] + \left[\left(\frac{-2220}{22.4\ \text{lit}}\right) 0.21 \quad \text{lit}\right]$$

$$\Delta H_C = \left[-614.66 \quad \text{kJ mol}^{-1}\right] + \left[-20.81 \quad \text{kJ mol}^{-1}\right]$$

$$\Delta H_C = 635.47\ \text{kJ mol}^{-1}$$

19.

$$4E_{C-H} = 360\ \text{kJ mol}^{-1}$$

$$E_{C-H} = 90\ \text{kJ mol}^{-1}$$

$$E_{C-C} + 6\ E_{C-H} = 620\ \text{kJ mol}^{-1}$$

$$E_{C-C} + 6 \times 90 = 620\ \text{kJ mol}^{-1}$$

$$E_{C-C} + 540 = 620\ \text{kJ mol}^{-1}$$

$$E_{C-C} = 80\ \text{kJ mol}^{-1}$$

22. During compression, energy of the system increases, in isothermal condition, to maintain temperature constant, heat is liberated from the system. Hence q is negative.

During compression entropy decreases.

During compression work is done on the system, hence w is positive

23.

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$T_b = \frac{\Delta H_V}{\Delta S_V} = \frac{4800\ \text{J mol}^{-1}}{16\ \text{J mol}^{-1}\ \text{K}^{-1}} = 300\ \text{K} = 27^0\,\text{C}$$

24. In $CaCO_3(S) \rightarrow CaO(S) + CO_2(g)$, entropy change is positive. In (a) and (b) entropy change is negative ; in (c) entropy change is zero.